

Ming-Hao Hsu

Ph.D. Student in Computer Science

☎ (+886) 979-990-106 | ✉ hsuminghao1006@gmail.com | 🏠 mhhsu1006.github.io | 📧 mhhsu1006

RESEARCH INTERESTS

Primary Spoken Language Models, Speech Tokenization, Mechanistic Interpretability
Secondary Text-to-Speech, Automatic Speech Recognition, Robot Learning, In-Context Learning
Methods Deep Learning, Self-Supervised Learning, Generative Models, Diffusion Models

EDUCATION

Chinese University of Hong Kong (CUHK), Shenzhen

Ph.D. Student in Computer Science

Shenzhen, China

Sep. 2025 – PRESENT

Tsinghua University (THU)

Exchange Program in Electrical Engineering

Beijing, China

Sep. 2023 – Jan. 2024

- GPA: 3.84/4.0

National Taiwan University (NTU)

B.S.E. in Electrical Engineering

Taipei, Taiwan

Sep. 2020 – Aug. 2024

- GPA: 3.88/4.3
- Honors: Presidential Award * 1 (F'20), which is given to top 5% students in class.
- Select Courses: Machine Learning 2022 Spring (A+), Computer Vision 2023 Spring(A+), Advanced Digital Signal Processing 2023 Spring (A+), Advanced Statistics I/II F'22/S'23 (A/A+)

RESEARCH EXPERIENCE

Microsoft Research Asia, Mentors: Dr. Jinyu Li, Dr. Shujie Liu

RESEARCH INTERN

Beijing, China

Jan. 2026 – PRESENT

- Researching on Speech and Language foundation models.

CUHK(SZ), Adviser: Prof. Zhizheng Wu

PH.D. STUDENT

Shenzhen, China

Sep. 2024 – PRESENT

- Researching on Paralinguistic Features for Spoken Language Models.
- Researching on Mechanistic Interpretability of Spoken Language Models, TTS models, and ASR models.
- Researching on Speech Tokenization for Spoken Language Models.

Tsinghua Embodied AI Lab, IIIS, THU, Adviser: Prof. Huazhe Xu

UNDERGRADUATE RESEARCHER

Beijing, China

Jul. 2023 – Oct. 2024

- Researched on Large Language Models for Robot Manipulation and Planning.
- Researched on utilizing Fail Trajectories for Imitation Learning.

- Researching on Spoken Language Models and Speech Tokenizers.
- Researched on utilizing Meta In-Context Learning on Whisper Model for low-resource language ASR. [[ASRU'25](#)]
- Researched on In-Context Learning for speech-language models. [[Interspeech'24](#)]
- Researched on Generative Language Models for Speech Question-Answering problems without speech recognition. [[Interspeech'24](#)]

Robot Learning Lab, NTU, Adviser: Prof. Shao-Hua Sun

UNDERGRADUATE RESEARCHER

Taipei, Taiwan

Sep. 2022 – Sep. 2024

- Researched on Imitation learning in Noisy and Partially Observable Environments.
- Researched on Diffusion Model Augmented Behavioral Cloning in various continuous control tasks. [[ICML'24](#)]

MiuLab, NTU, Adviser: Prof. Yun-Nung (Vivian) Chen

UNDERGRADUATE RESEARCHER

Taipei, Taiwan

Sep. 2021 – Sep. 2022

- Researched on Data Augmentation Model for dialogue state tracking system. (National Science and Technology Council Research project, No. 111-2813-C-002-083-E) [[SIGDIAL'22](#)]

PUBLICATIONS

- [1] Ryan Soh-Eun Shim, Kwanghee Choi, Calvin Chang, Ming-Hao Hsu, Florian Eichin, Zhizheng Wu, Alane Suhr, Michael A. Hedderich, David Harwath, David R. Mortensen, and Barbara Plank. [Linear Script Representations in Speech Foundation Models Enable Zero-Shot Transliteration](#). In *Arxiv Preprint*, 2026.
- [2] Ming-Hao Hsu, Liang-Hsuan Tseng, Hung-yi Lee, and Zhizheng Wu. [TASLA: Text-Aligned Speech Tokens with Multiple Layer-Aggregation](#). In *Arxiv Preprint*, 2025.
- [3] Ming-Hao Hsu and Hung-yi Lee. [SMILE: Speech Meta In-Context Learning for Low-Resource Language ASR](#). In *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)* [Best Student Paper Finalist], 2025.
- [4] Shang-Fu Chen*, Hsiang-Chun Wang*, Ming-Hao Hsu, Chun-Mao Lai, and Shao-Hua Sun. [Diffusion Model-Augmented Behavioral Cloning](#). In *Forty-first International Conference on Machine Learning (ICML)*, 2024.
- [5] Ming-Hao Hsu*, Kai-Wei Chang*, Shang-Wen Li, and Hung yi Lee. [Exploring In-Context Learning of Textless Speech Language Model for Speech Classification Tasks](#). In *Proceedings of the 2024 Conference of the International Speech Communication Association (INTERSPEECH)*, 2024.
- [6] Min-Han Shih*, Ho-Lam Chung*, Yu-Chi Pai*, Ming-Hao Hsu, Guan-Ting Lin, Shang-Wen Li, and Hung yi Lee. [GSQA: An End-to-End Model for Generative Spoken Question Answering](#). In *Proceedings of the 2024 Conference of the International Speech Communication Association (INTERSPEECH)*, 2024.
- [7] Chun-Mao Lai*, Ming-Hao Hsu*, Chao-Wei Huang, and Yun-Nung Chen. [Controllable User Dialogue Act Augmentation for Dialogue State Tracking](#). In *Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue, SIGDIAL 2022, Edinburgh, UK, 07-09 September 2022*, pages 53–61. Association for Computational Linguistics, 2022.

HONORS & AWARDS

2025	Best Student Paper Finalist – IEEE ASRU 2025	Hawaii, USA
2024	Final List – Legal Compliance Tech Hackathon, Judiciary of Taiwan	Taipei, Taiwan
2024	Award Winner – NTUEE Ma Tso-Ping Academician Future Innovation Leaders Scholarship	Taipei, Taiwan
2023	3rd Place – NTUEE Undergraduate Innovation Award	Taipei, Taiwan
2022	Finalist – NTU Hult Prize	Taipei, Taiwan
2021	Presidential Award 20'Fall – NTU	Taipei, Taiwan

PROFESSIONAL SERVICES

Conference Reviewer: NeurIPS 2024, ICLR 2025, ICASSP 2025
Journal Reviewer: IEEE Open Journal of Signal Processing